

1. Administrative & Study Identification

Full Study Title: Phase III Study of Efficacy and Safety of Secukinumab Versus Placebo, in Combination With Glucocorticoid Taper Regimen, in Patients With Giant Cell Arteritis (GCA) **Short Title/Acronym:** GCaptAIN **Protocol Number:** CAIN457R12301 **NCT Number:** NCT04930094 **Trial Status:** Active, not recruiting **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Secukinumab (Cosentyx / AIN457) | **ATC Code:** L04AC10 **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess whether secukinumab 300 mg subcutaneously (s.c.) plus a 26-week glucocorticoid (GC) taper is superior to placebo plus a 52-week GC taper in achieving sustained remission at Week 52. **Secondary Objectives:** To evaluate the cumulative GC dose through Week 52, remission rate at Week 12, time to first GCA disease flare, patient-reported outcomes (PROs), and overall safety and tolerability.

2.2 Background & Rationale Giant cell arteritis (GCA) is the most common form of systemic vasculitis in adults over 50. It can cause irreversible vision loss and aortic aneurysms, making it a medical emergency. Current standard treatment relies heavily on long-term glucocorticoids, which cause significant steroid-related toxicity. Secukinumab, an IL-17A inhibitor, targets the underlying inflammatory pathways and is being evaluated to maintain sustained remission while significantly reducing the required cumulative steroid dose.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1:1 (Secukinumab 300 mg : Secukinumab 150 mg : Placebo)

3.2 Planned Enrollment & Duration **Trial Status:** Active, not recruiting **Target \$N\$:** 355 patients **Number of Sites:** 104 locations across 27 countries **Estimated Study Start:** 15-Jun-2021 **Estimated Primary Completion:** Mid-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or non-pregnant, non-lactating female patients at least 50 years of age. 2. Diagnosis of GCA based on age at onset ≥ 50 years and unequivocal cranial symptoms of GCA (e.g., new-onset localized headache). 3. Patients meeting the definition of new-onset GCA or relapsing GCA within 6 weeks of Baseline (BSL).

4.2 Exclusion Criteria 1. Pregnant or nursing (lactating) women. 2. Previous exposure to secukinumab or other biologic drugs directly targeting IL-17 or IL-17 receptor. 3. Significant medical problems including uncontrolled hypertension ($\geq 160/95$ mmHg), severe congestive heart failure (NYHA class III or IV), or uncontrolled diabetes mellitus.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Secukinumab 300 mg or 150 mg s.c. administered at Baseline, Weeks 1, 2, 3, and 4, followed by administration every 4 weeks. **Route of Administration:** Subcutaneous (s.c.) **Duration of Treatment:** 52 weeks

5.2 Concomitant & Prohibited Therapy Permitted: Protocol-specified 26-week prednisone (glucocorticoid) taper regimen for the active arm, and 52-week taper for the placebo arm. **Prohibited:** Cell-depleting therapies, inhibitors targeting IL-1, IL-12/23, or abatacept within 4 weeks of baseline.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to -42 Days	Informed Consent, Medical History, Diagnosis Confirmation, Labs
Baseline	Day 0	Randomization, First Dose of Secukinumab/Placebo, Initiation of GC Taper
Interim	Weeks 1, 2, 3, 4, then Q4W	Subcutaneous Injections, Efficacy & Flare Assessment, Steroid Taper Tracking
End of Study	Week 52	Final Efficacy Assessment, Sustained Remission Evaluation, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of GCA patients achieving sustained remission at Week 52. **Measurement Method:** Clinical evaluation of GCA signs and symptoms without the need for rescue therapy, alongside monitoring of erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP).

7.2 Secondary & Exploratory Endpoints 1. Cumulative glucocorticoid (GC) dose through Week 52. 2. Time to first GCA disease flare. 3. Remission rate at Week 12.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via patient reporting, laboratory tests, and clinical evaluation at every 4-week visit. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor; specific monitoring for infections and steroid-associated toxicities. **Data Monitoring Committee (DMC):** Independent external committee responsible for periodic safety and efficacy reviews.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for all randomized patients receiving at least one dose. **Sample Size Justification:** Powered to detect a statistically significant difference in the proportion of patients achieving sustained remission at Week 52 comparing Secukinumab 300 mg vs. Placebo. **Handling of Missing Data:** Non-responder imputation (NRI) for binary primary endpoints; multiple imputation or mixed-effects model for repeated measures (MMRM) for continuous variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence and accurate GC dose tracking. **Audit Trail:** Secure eCRF with full audit capabilities and electronic data clarification forms.

11. Study Identifiers & Governance

Primary ID: CAIN457R12301 **Registry Number:** NCT04930094 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** GCaptAIN **Official Regulatory Title:** A Randomized, Parallel-group, Double-blind, Placebo-controlled, Multicenter Phase III Trial to Investigate the Efficacy and Safety of Secukinumab 300 mg and 150 mg Administered Subcutaneously Versus Placebo, in Combination With a Glucocorticoid Taper Regimen, in Patients With Giant Cell Arteritis (GCA) (GCaptAIN) **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/cosentyx-epar-product-information_en.pdf

12. Clinical Framework (Giant Cell Arteritis Specific)

Primary Condition: Giant Cell Arteritis (GCA) **Preferred UMLS Name:** Giant Cell Arteritis **Semantic Type:** Disease or Syndrome **MeSH Code:** D013700 | **SNOMED ID:** 400130008 **Diagnosis Threshold:** Age $\geq$ 50 years, active cranial symptoms, and confirmed new-onset or relapsing GCA **Medical Methodology:** IL-17A inhibition combined with Glucocorticoid (GC) Taper Regimen **Optimization Target:** Sustained remission and cumulative steroid dose reduction

13. Study Procedures & Methodology

Management Pathway: Protocol-specified 26-week prednisone taper alongside Secukinumab administration **Education Protocol (HCP):** Training on standardized GC tapering, recognizing GCA flares, and AE monitoring **Education Protocol (Patient):** Subcutaneous self-injection training, symptom monitoring (headaches, vision changes), and adherence **Quality Audit Mechanism:** Tracking of exact glucocorticoid doses utilized, patient diaries, and central lab monitoring of ESR/CRP

14. Observation & Follow-Up Schedule

Total Duration: 52 Weeks of treatment (plus safety follow-up) **Onsite Visit Cadence:** Baseline, Weeks 1, 2, 3, 4, and every 4 weeks thereafter up to Week 52 **Remote Monitoring Frequency:** Between onsite visits as clinically indicated **Checklist Review Interval:** Every 4 weeks (Q4W) to evaluate remission status and taper adherence

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				